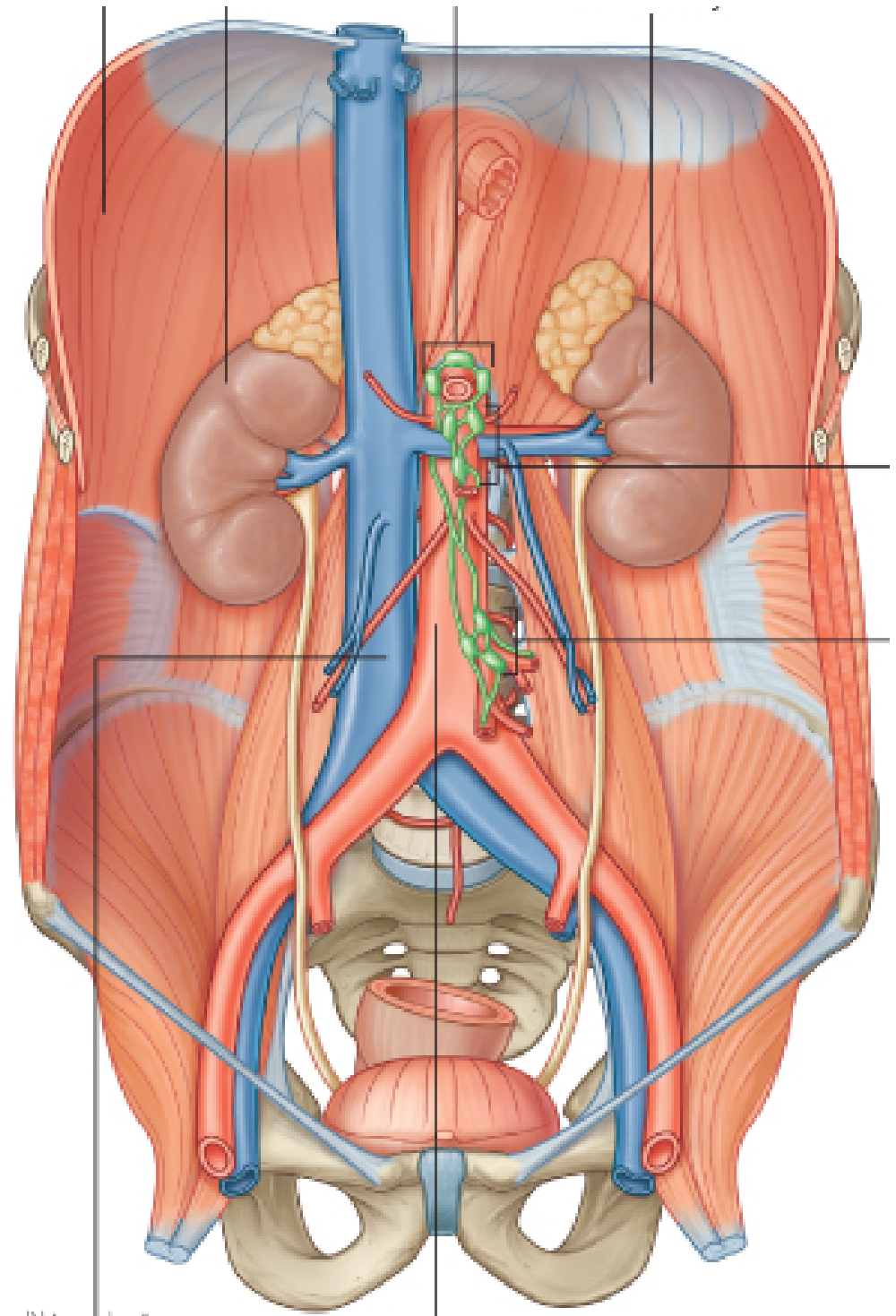


Anatomy of The Kidney

BY: M. G. HAMMAD



Anatomy of The Kidney



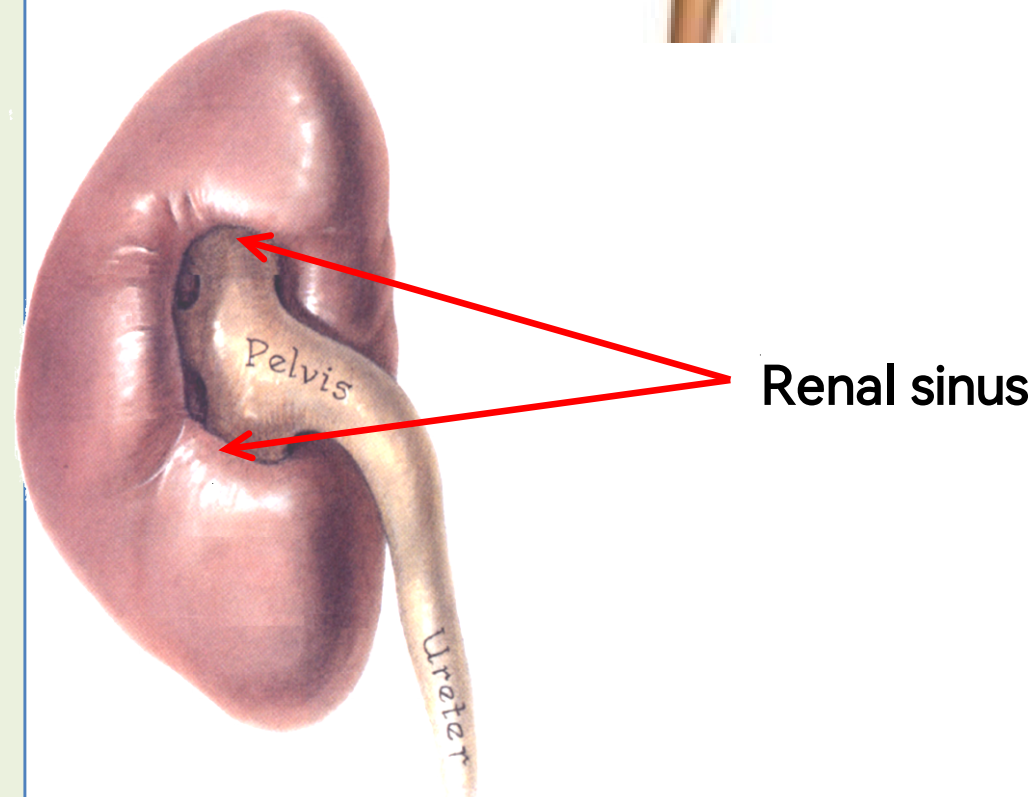
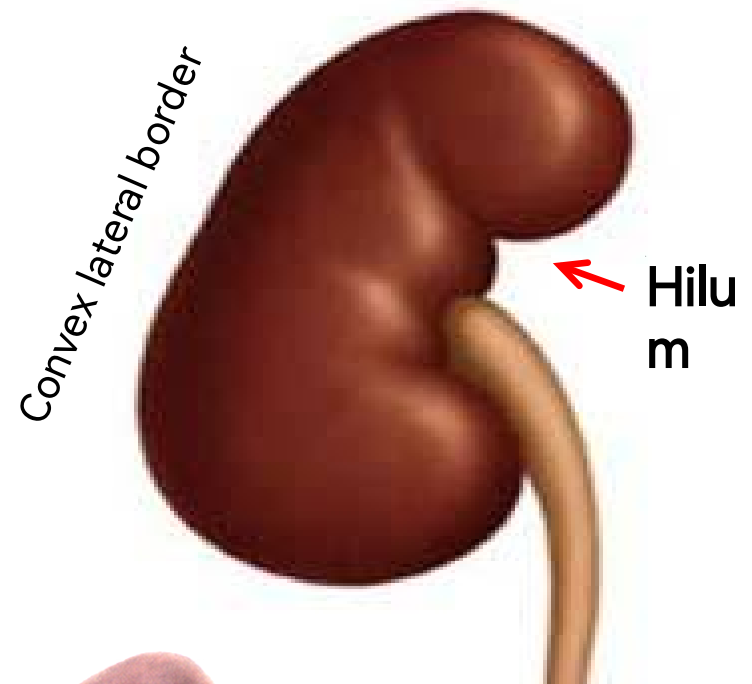
Objectives

- ❑ By the end of the lecture, the student should be able to describe the:
 - Anatomical features of the kidneys:
 - ✓ Position, extent, relations, hilum, peritoneal coverings.
 - Internal structure of the kidneys:
 - ✓ Cortex, medulla and renal sinus.
 - The vascular segments of the kidneys.
 - The blood supply and lymphatics of the kidneys.



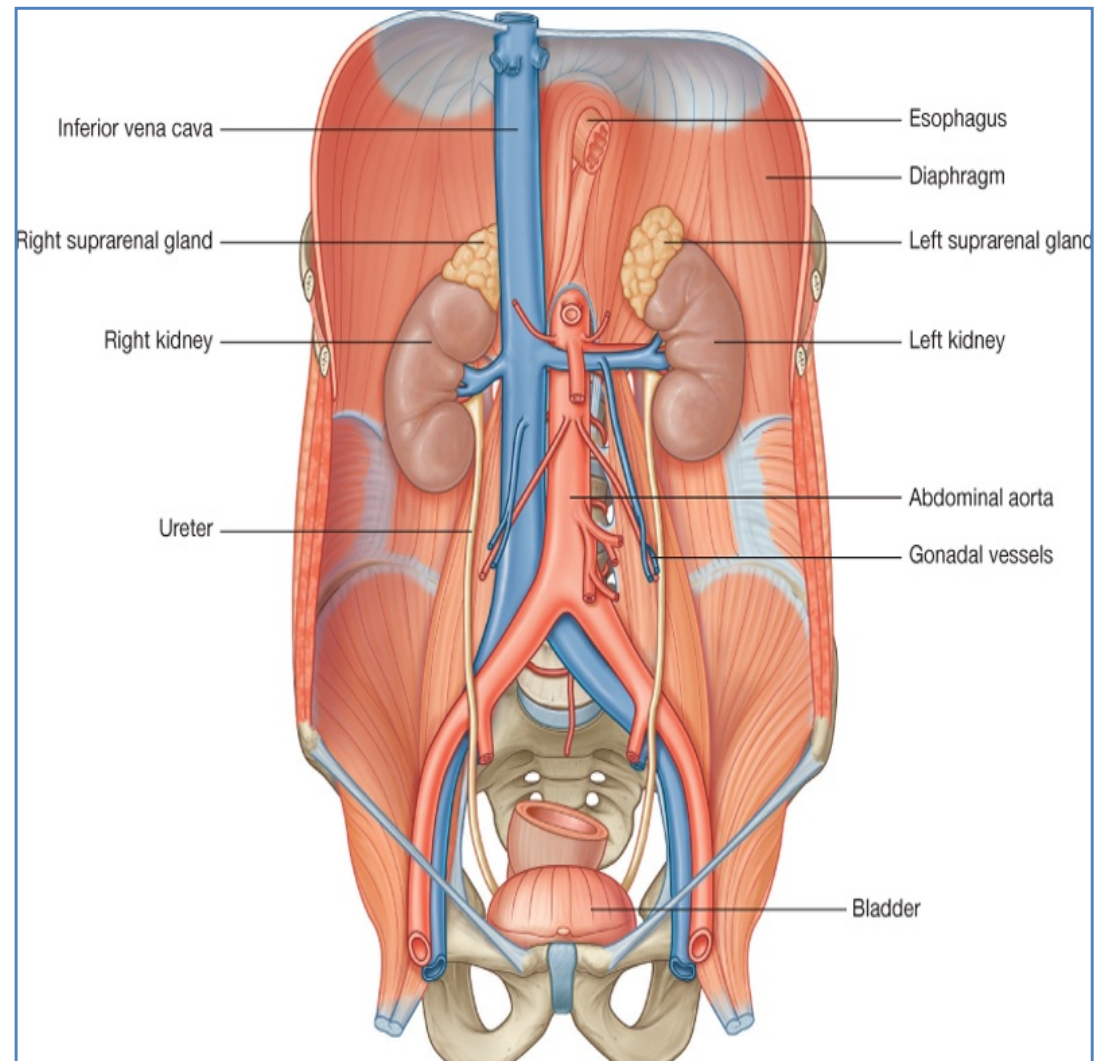
Color, Shape & Dimensions

- The kidney is a reddish brown, bean-shaped organ with the dimensions 12 x 6 x 3cm.
- Left kidney is slightly longer and more slender than the right kidney, and nearer to the midline.
- Each kidney has:
 - Convex upper & lower ends.
 - Convex lateral border.
 - Convex medial border at both ends, but its middle shows a vertical slit called the hilum.
- Internally the hilum extends into a large cavity called the renal sinus.



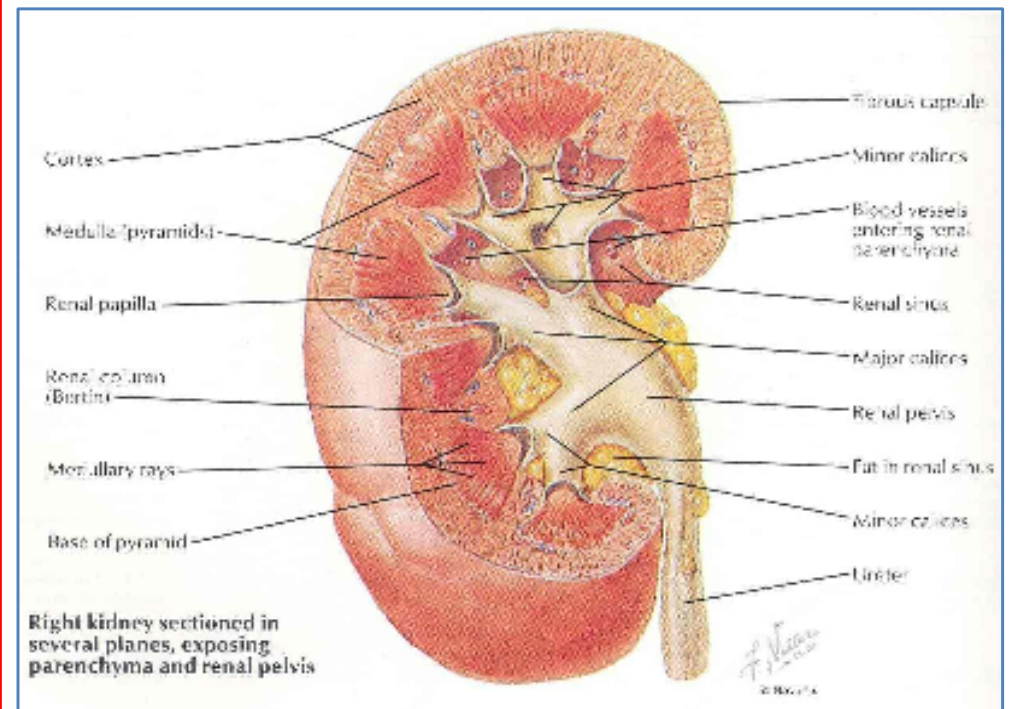
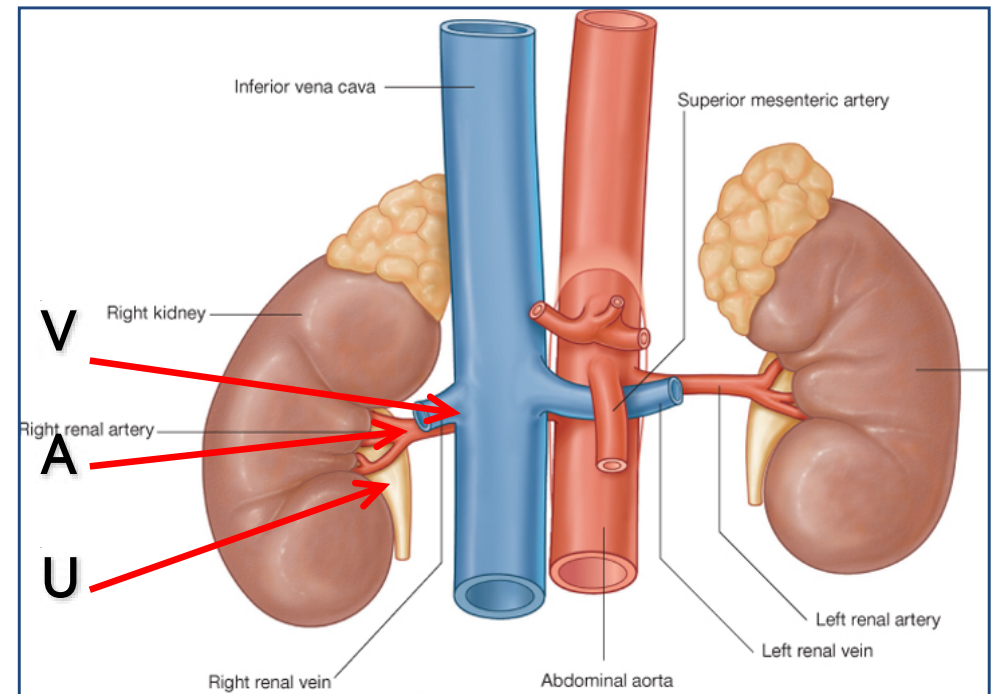
Position of the kidneys

- Kidneys are retroperitoneal paired organs.
- Lies on the posterior abdominal wall.
- Lateral to the vertebral column.
- In the supine position, the kidneys extend from T12 to L3.
- The right kidney is slightly lower than the left kidney because of the large size of the right lobe of the liver.
- With contraction of the diaphragm during respiration, both kidneys move downward in a vertical direction (high of one vertebra, 1 inch, 2.5



Hilum & Renal sinus

- The hilum transmits, from anterior to posterior, the renal vein, renal artery & the ureter (VAU).
- Lymph vessels & sympathetic fibers also pass through the hilum.
- The renal sinus contains the upper expanded part of the ureter called the renal pelvis.
- Perinephric fat is continuous into the hilum and the sinus and surrounds all these structures.



Coverings

❑ Fibrous capsule:

- ✓ Is closely adherent to its surface

❑ Perirenal fat:

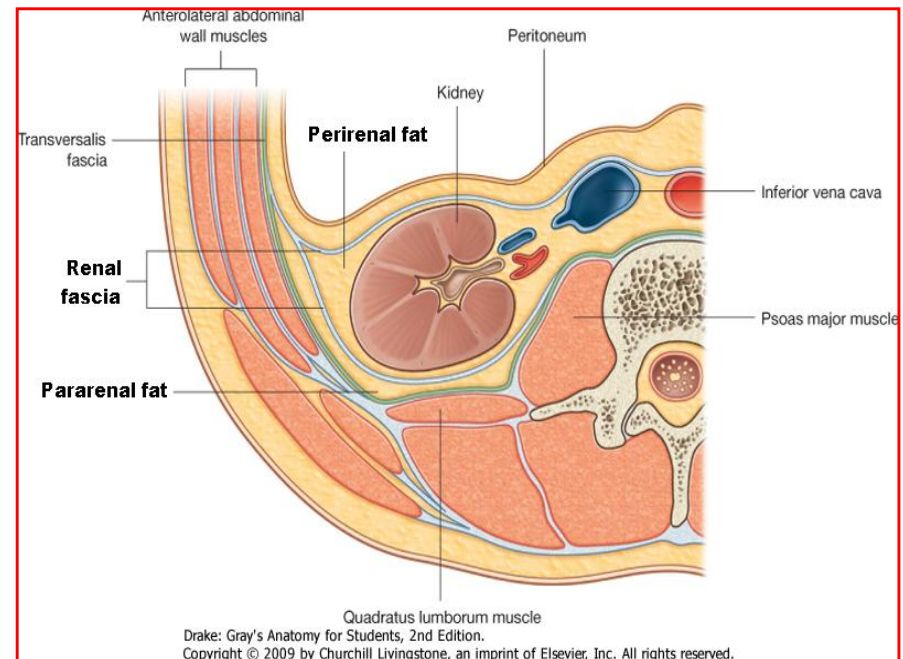
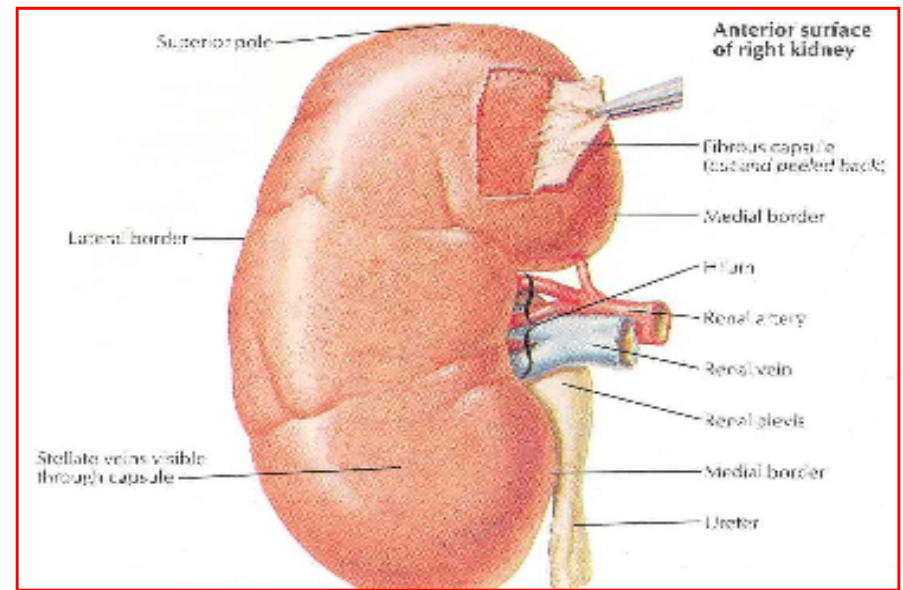
- ✓ Covers the fibrous capsule.

❑ Renal fascia:

- ✓ Condensation of areolar connective tissue that lies outside the Perirenal fat and encloses the kidney and the suprarenal gland.

❑ Pararenal fat:

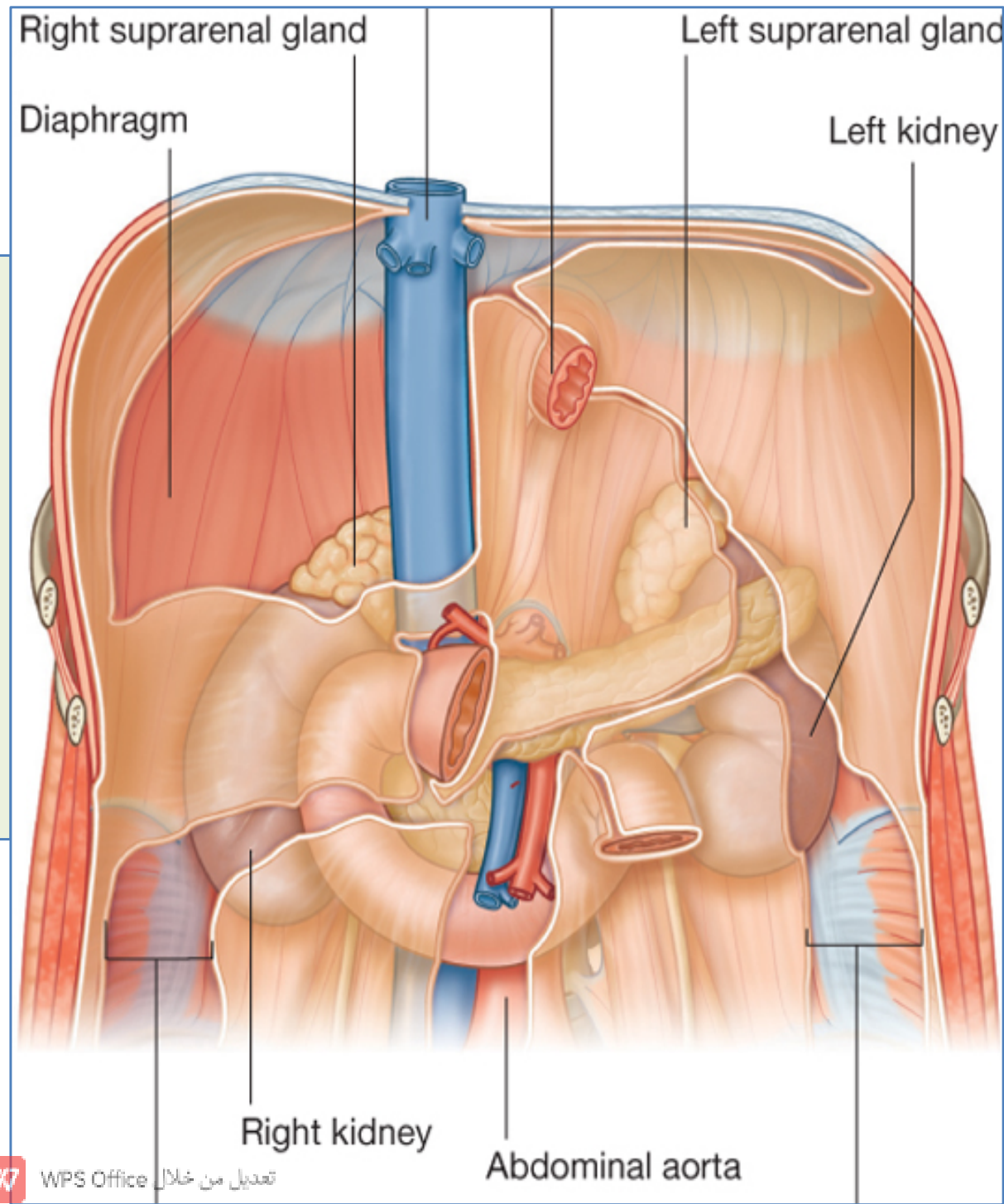
- ✓ Lies external to the renal fascia, is part of the retroperitoneal fat.



- The last 3 structures support the kidneys and hold it in position on the posterior abdominal wall.

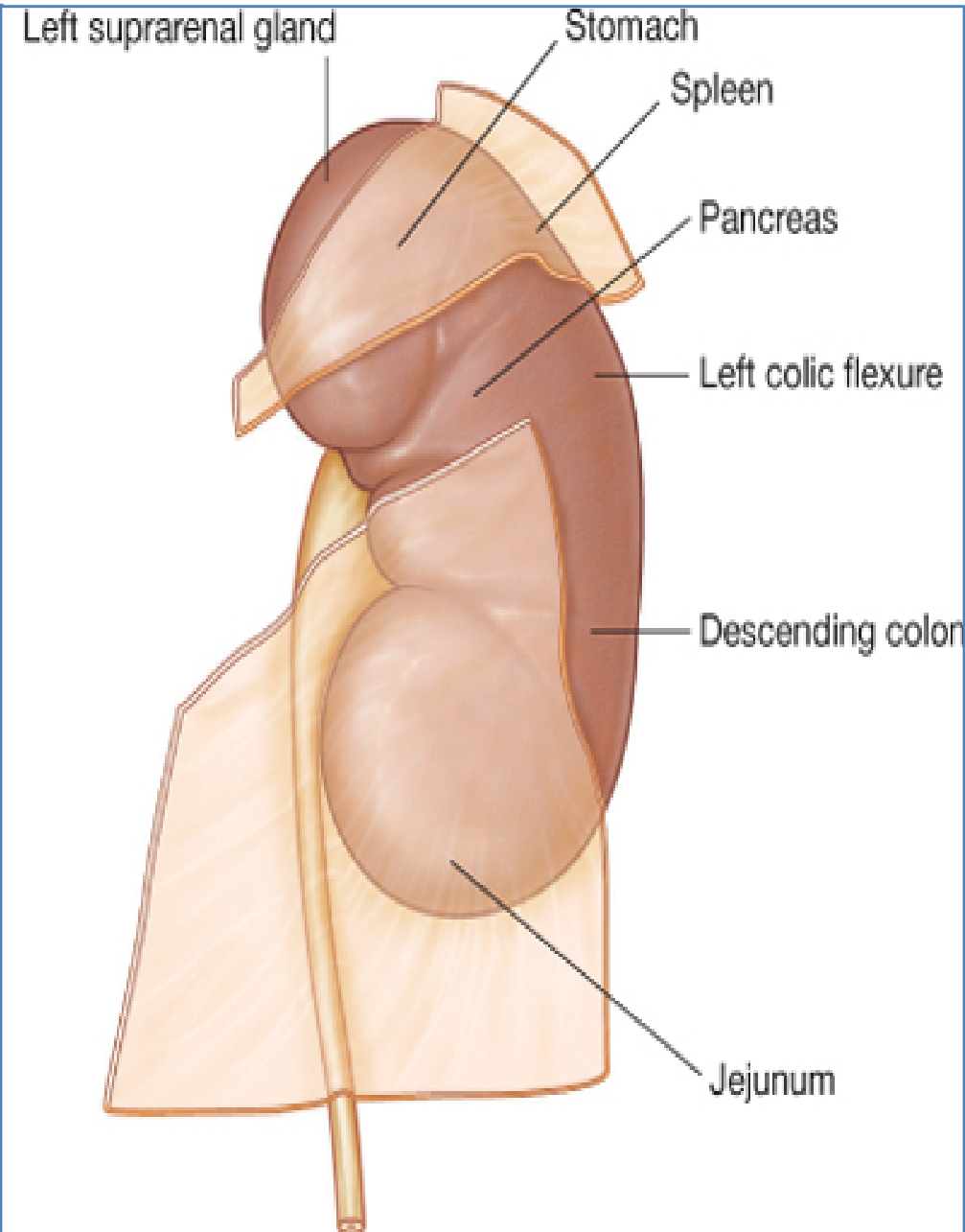
Relations Anterior

- ❑ The anterior surface of both kidneys are related to numerous structures.
- ❑ Some with intervening layer of peritoneum.
- ❑ Others lie directly against the kidney without peritoneum.



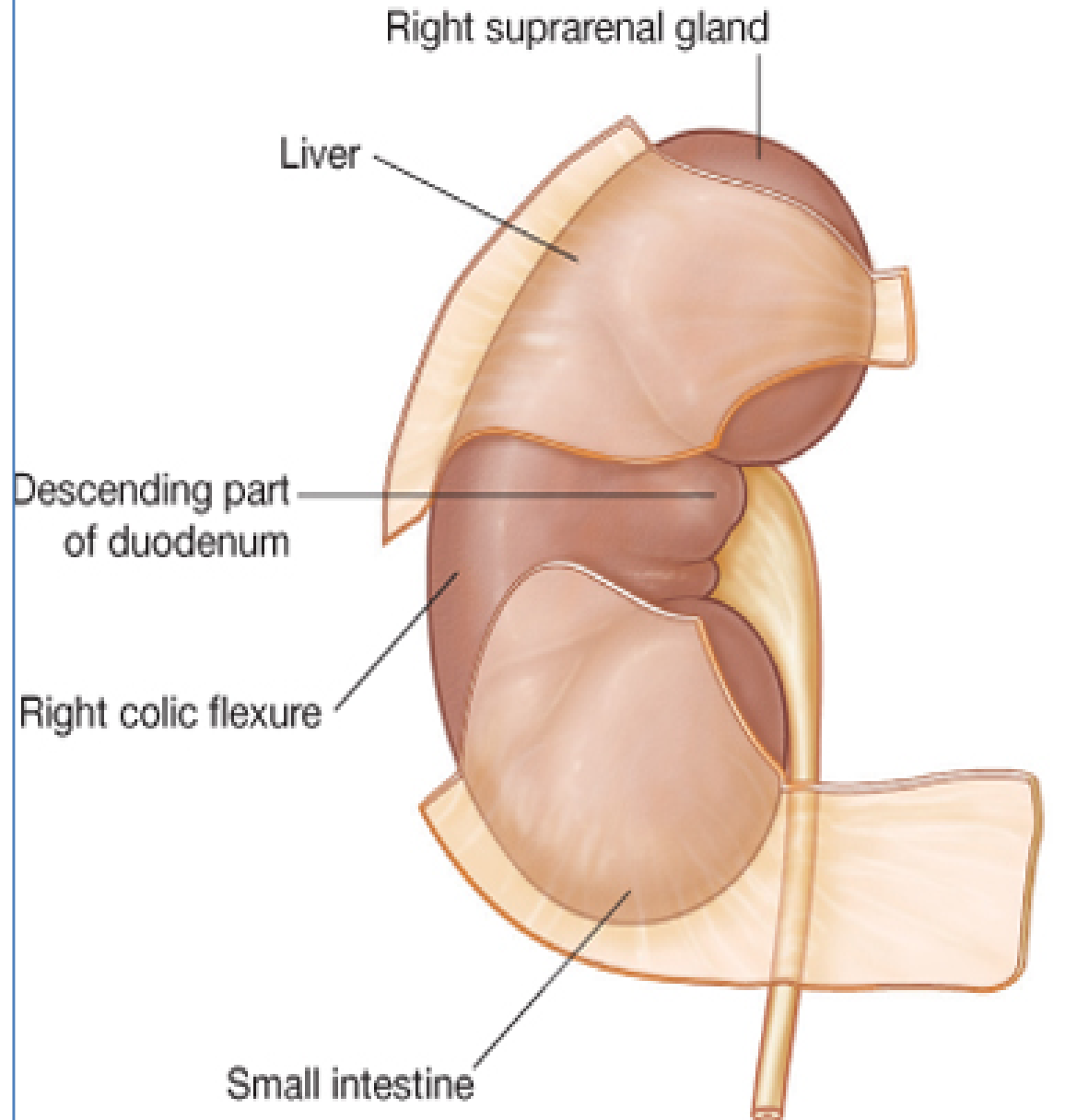
❑ Left kidney:

- ✓ A small part of the superior pole, along the medial border, is covered by left suprarenal gland.
- ✓ The rest of the superior pole is covered by the intraperitoneal stomach and spleen.
- ✓ The retroperitoneal pancreas covers the middle part of the kidney.
- ✓ Its lower lateral part is directly related to the left colic flexure and beginning of descending colon.
- ✓ Its lower medial part is covered by the intraperitoneal jejunum.



❑ Right kidney:

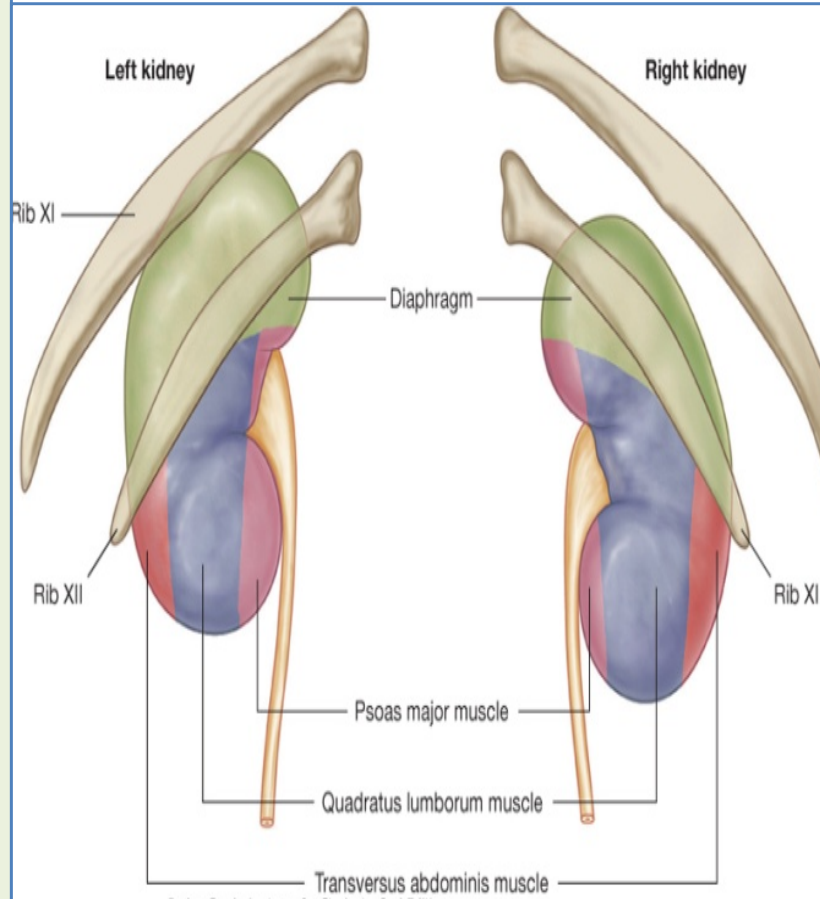
- A small part of the upper pole is covered by right suprarenal gland.
- The rest of the upper part of anterior surface is related to the liver and is separated by a layer of peritoneum.
- The 2nd part of duodenum lies directly in front of the kidney close to its hilum.
- The lower lateral part is directly related to the right colic flexure and, on its lower medial side, is related to the intraperitoneal small intestine.



Posterior Relations

Posteriorly, the right and left kidneys are almost related to similar structures.

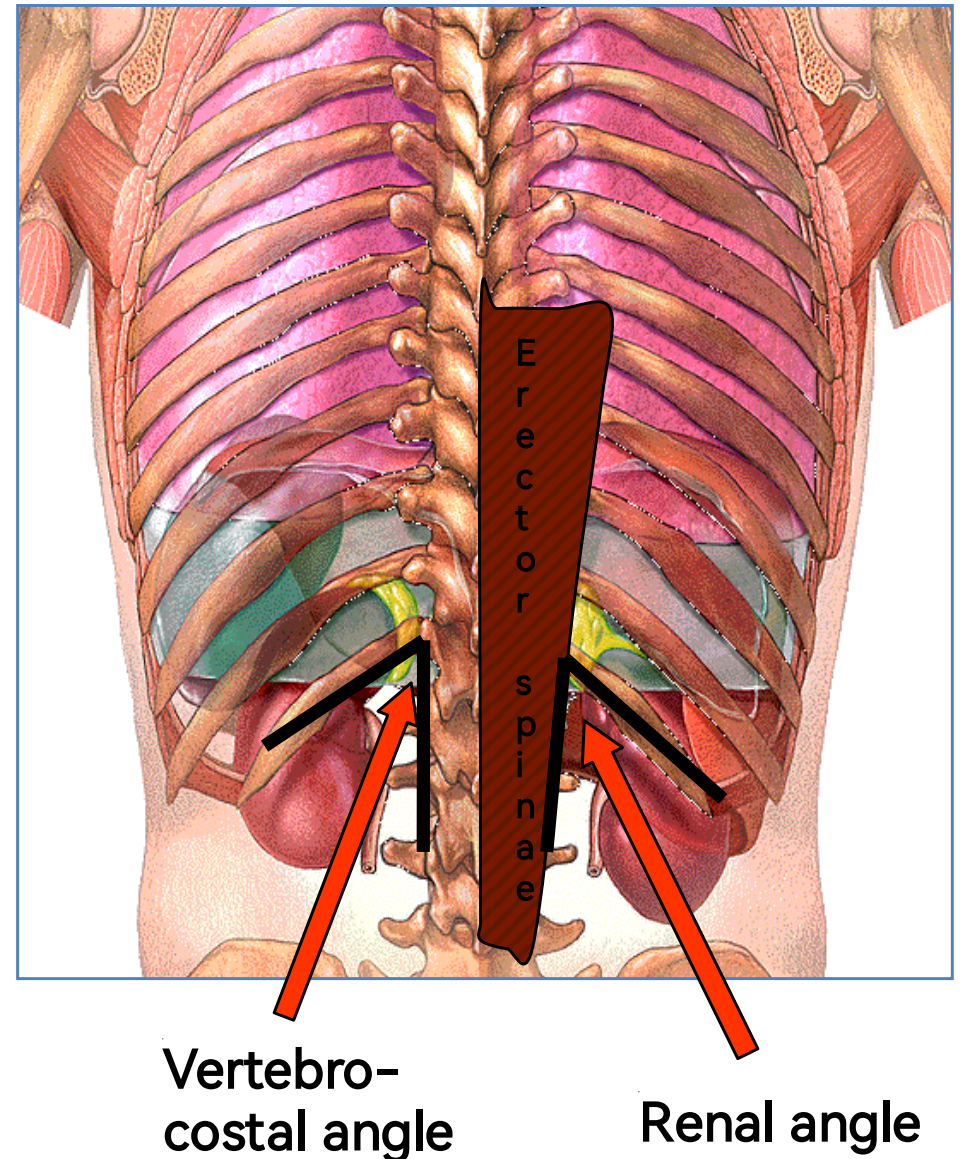
- ❑ **Left kidney:**
- ✓ Diaphragm
- ✓ Costodiaphragmatic recess of pleura.
- ✓ 11th & 12th ribs, last intercostal space.
- ✓ Psoas major
- ✓ Quadratus lumborum.
- ✓ Transversus abdominis.
- ✓ Subcostal, iliohypogastric & ilioinguinal nerves.



- ❑ **Right kidney:**
- ✓ Diaphragm
- ✓ Costodiaphragmatic recess of the pleura.
- ✓ 12th rib, last intercostal space.
- ✓ Psoas major
- ✓ Quadratus lumborum.
- ✓ Transversus abdominis.
- ✓ Subcostal, iliohypogastric & ilioinguinal nerves.

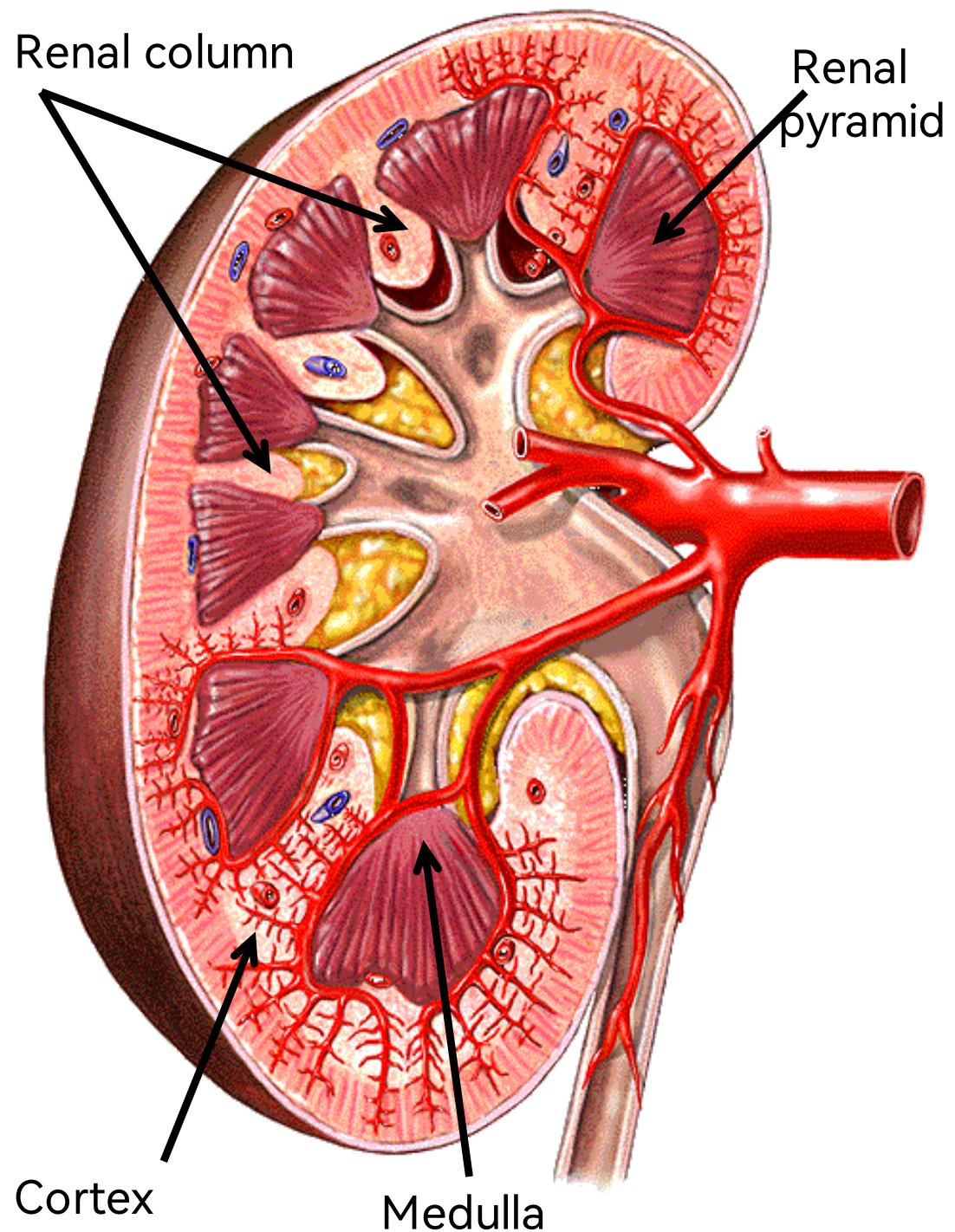
Vertebrocostal & Renal Angles

- ✓ The angle between the last rib and the lateral border of erector spinae muscle is occupied by kidney and is called the 'Renal angle'
- ✓ The Vertebrocostal angle is occupied by the lower part of the pleural sac.

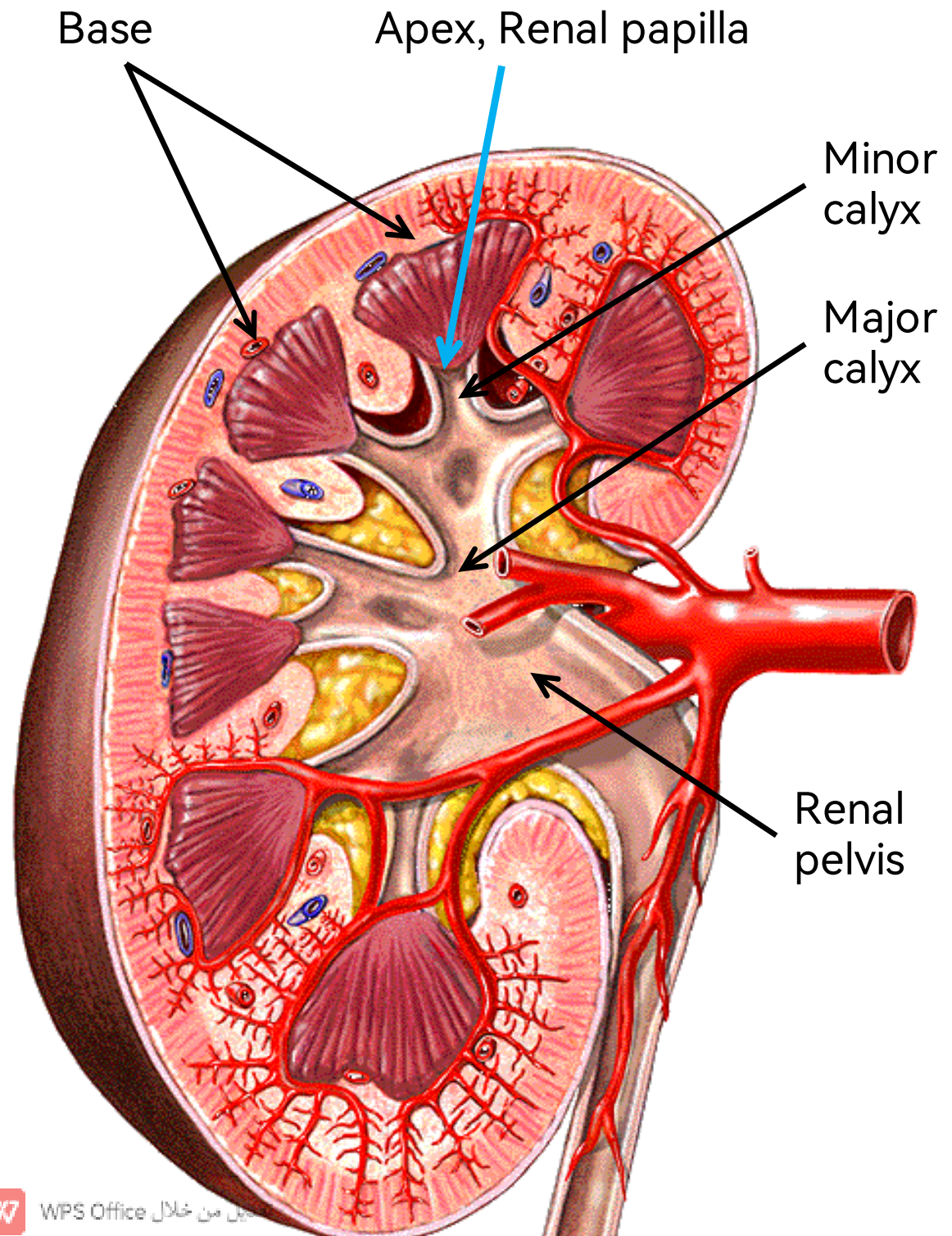


Internal structure

- ✓ Each kidney consists of an outer renal cortex and an inner renal medulla.
- ✓ The renal cortex is a continuous band of pale tissue that completely surrounds the renal medulla.
- ✓ Extensions of the renal cortex, the renal columns project into the inner aspect of the kidney, dividing the renal medulla into discontinuous aggregations of triangular-shaped tissue, the renal pyramids.

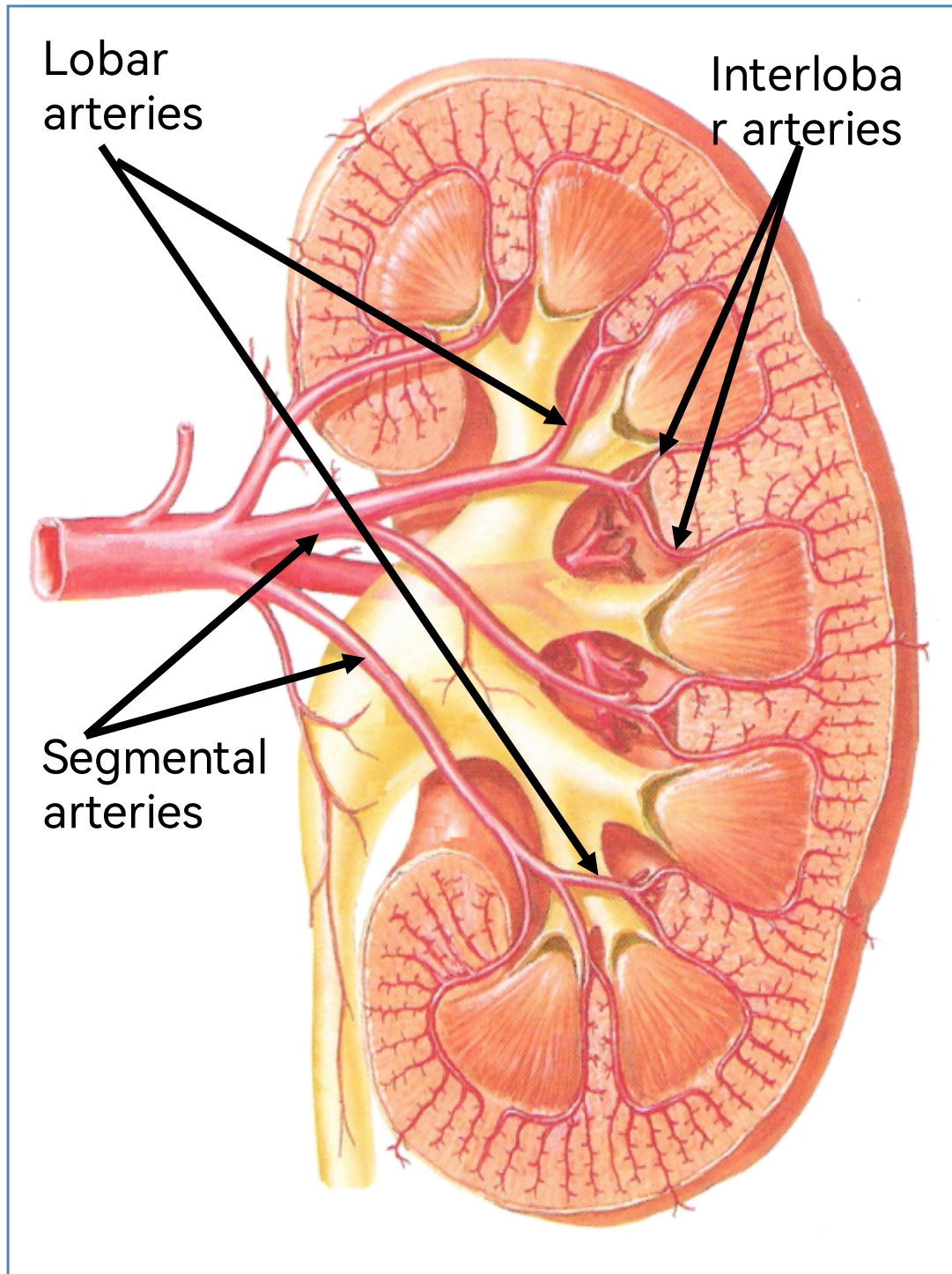


- The bases of the renal pyramids are directed outward, toward the cortex, while the apex of each renal pyramid projects inward, toward the renal sinus.
- The apical projection (renal papilla) is surrounded by a minor calyx.
- In the renal sinus, several minor calices unite to form a major calyx, and two or three major calices unite to form the renal pelvis, which is the funnel-shaped superior end of the ureters.

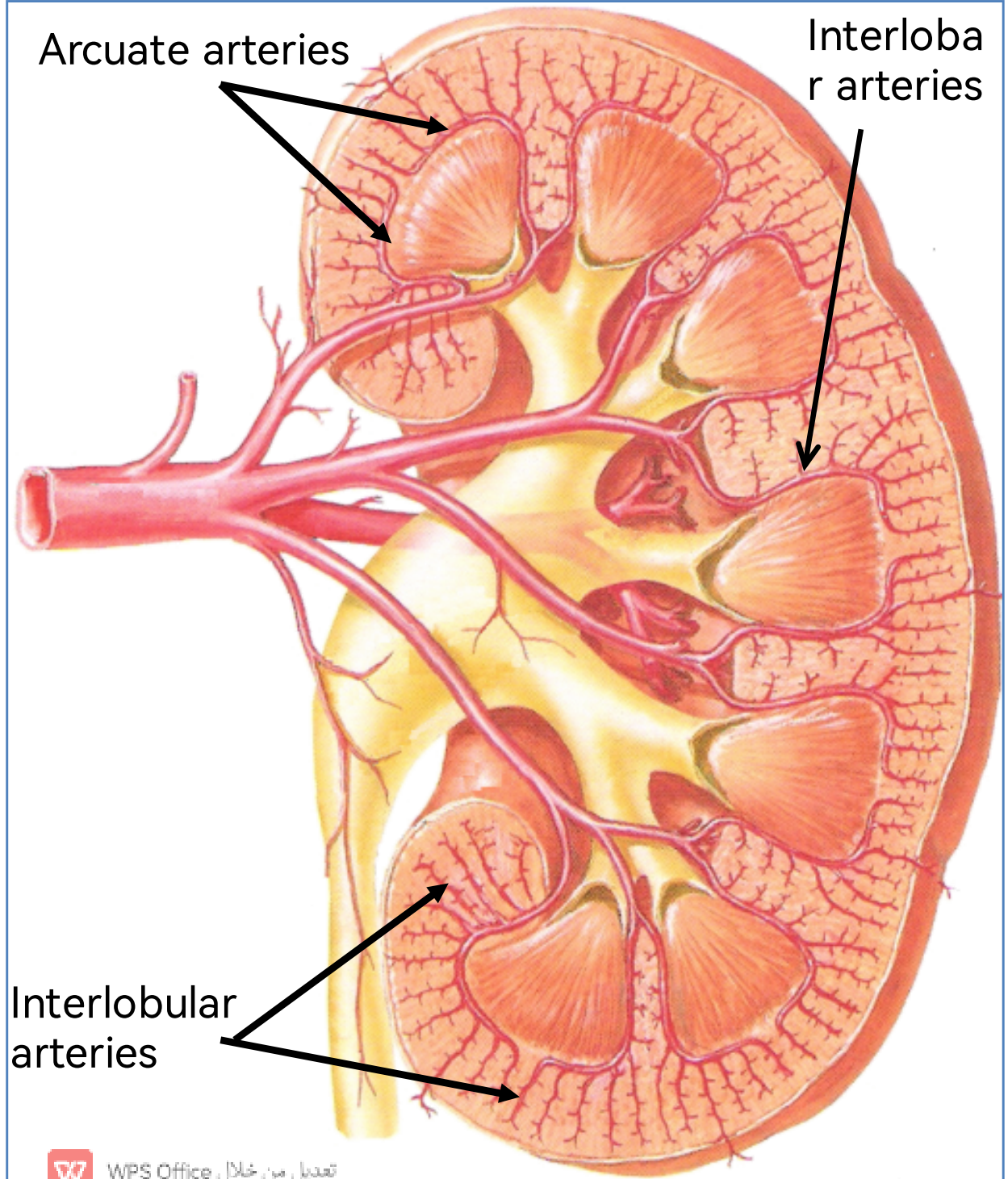


Arterial Supply

- The renal artery arises from the aorta at the level of the second lumbar vertebra.
- Each renal artery divides into 5 segmental arteries that enter the hilum of the kidney, 4 in front of the renal pelvis and one behind it.
- They are distributed to the different segments of the kidney.
- Each segmental artery gives rise to number of lobar arteries, each supplies a renal pyramid.
- Before entering the renal substance, each lobar artery gives off two or three interlobar arteries.



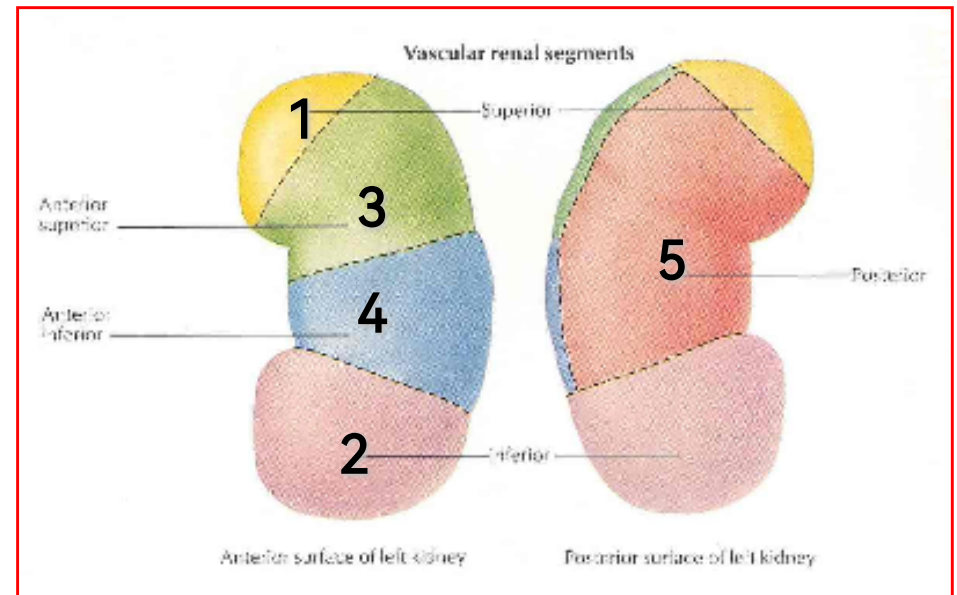
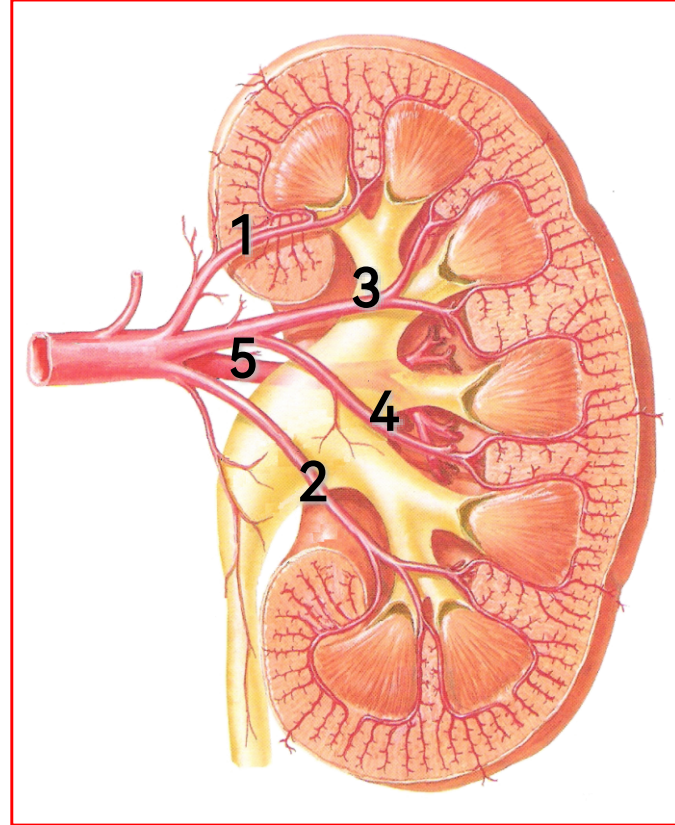
- The interlobar arteries run toward the cortex on each side of the renal pyramid.
- At the junction of the cortex and the medulla, the Interlobar arteries give off the arcuate arteries, which arch over the bases of the pyramids.
- The arcuate arteries give off several interlobular arteries that ascend in the cortex and give off the afferent glomerular arterioles.



Segmental branches & vascular segments of kidneys

- Each kidney has 5 segmental branches and is divided into 5 vascular segments:

1. Apical.
2. Caudal.
3. Anterior Superior.
4. Anterior Inferior.
5. Posterior.



Blood Supply

- Abdominal aorta



- Renal artery



- Segmental arteries



- lobar arteries



- Interlobar arteries



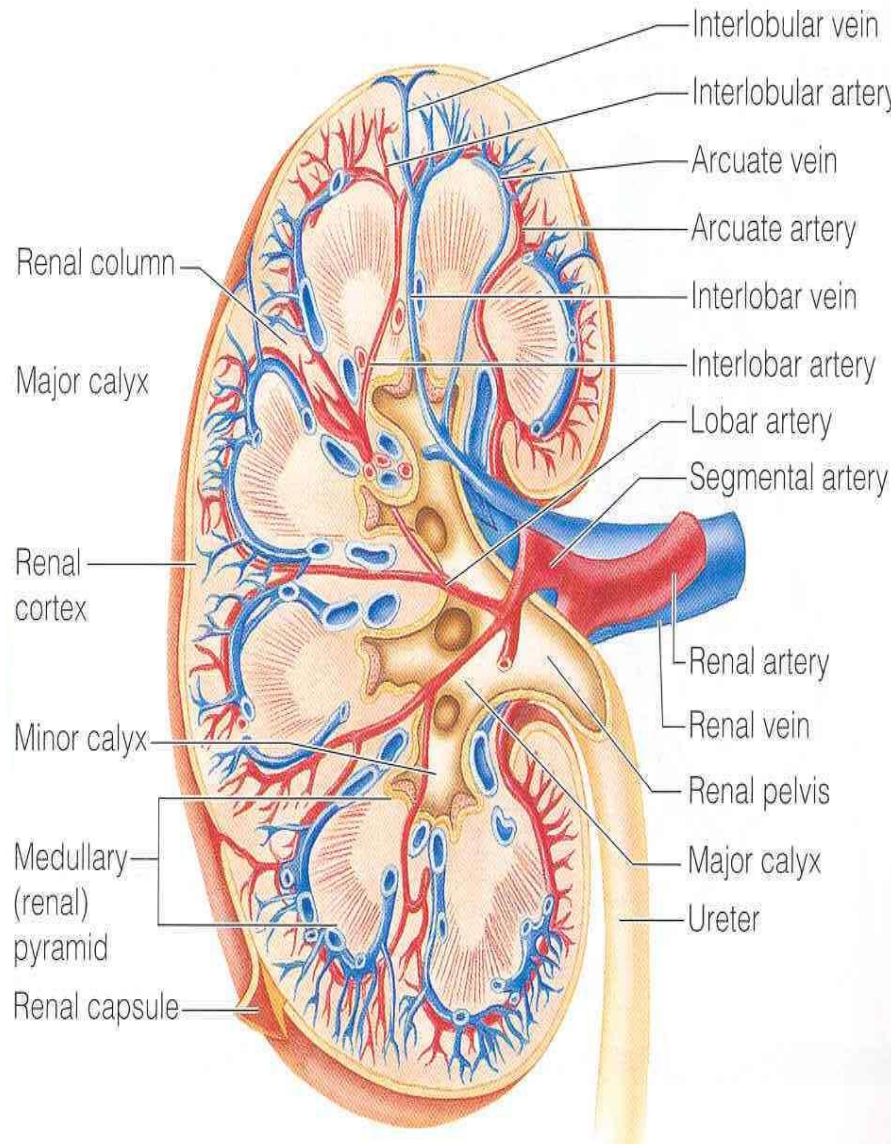
- Arcuate arteries



- Interlobular arteries



- afferent



- Inferior vena cava



- Renal vein



- Interlobar veins



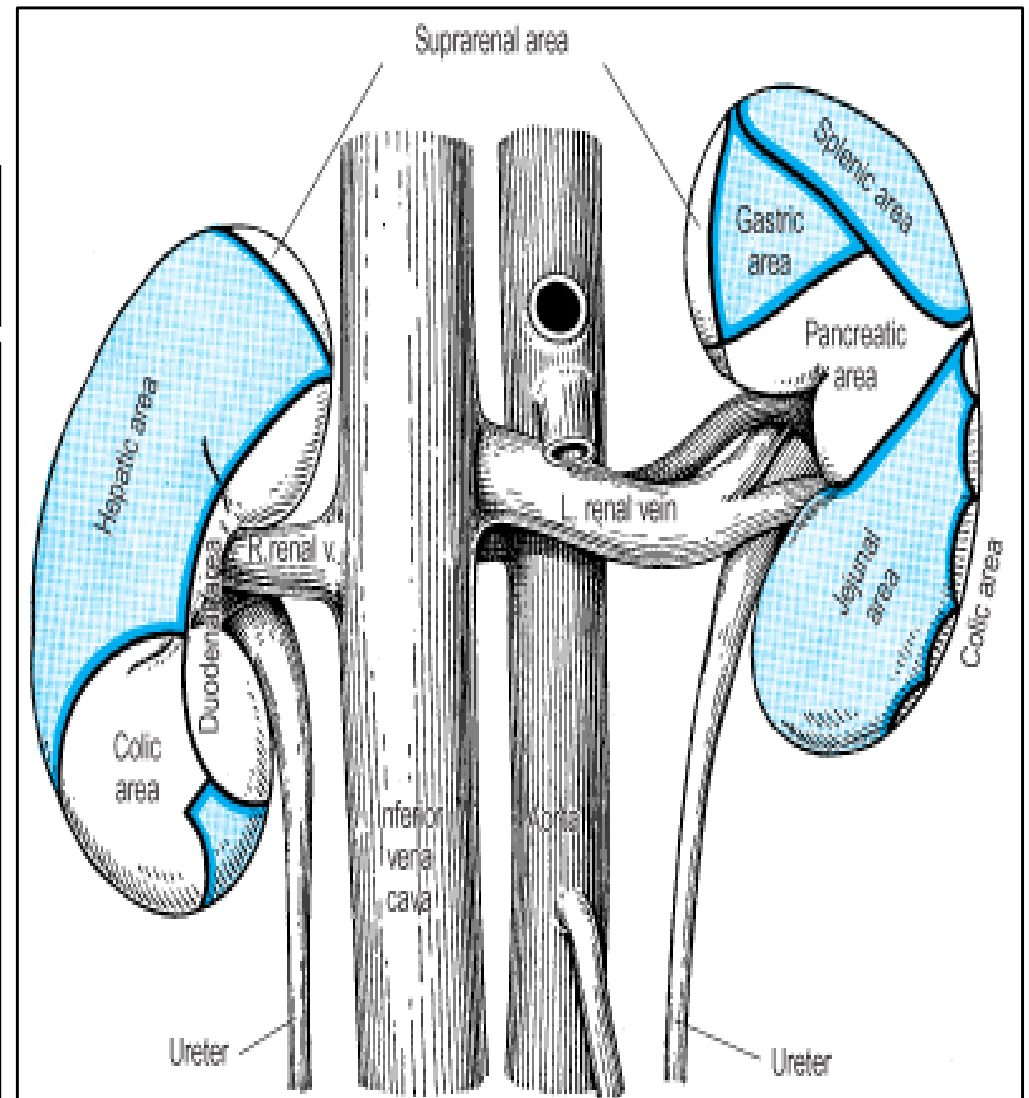
- Arcuate veins



- Interlobular veins

Venous Drainage

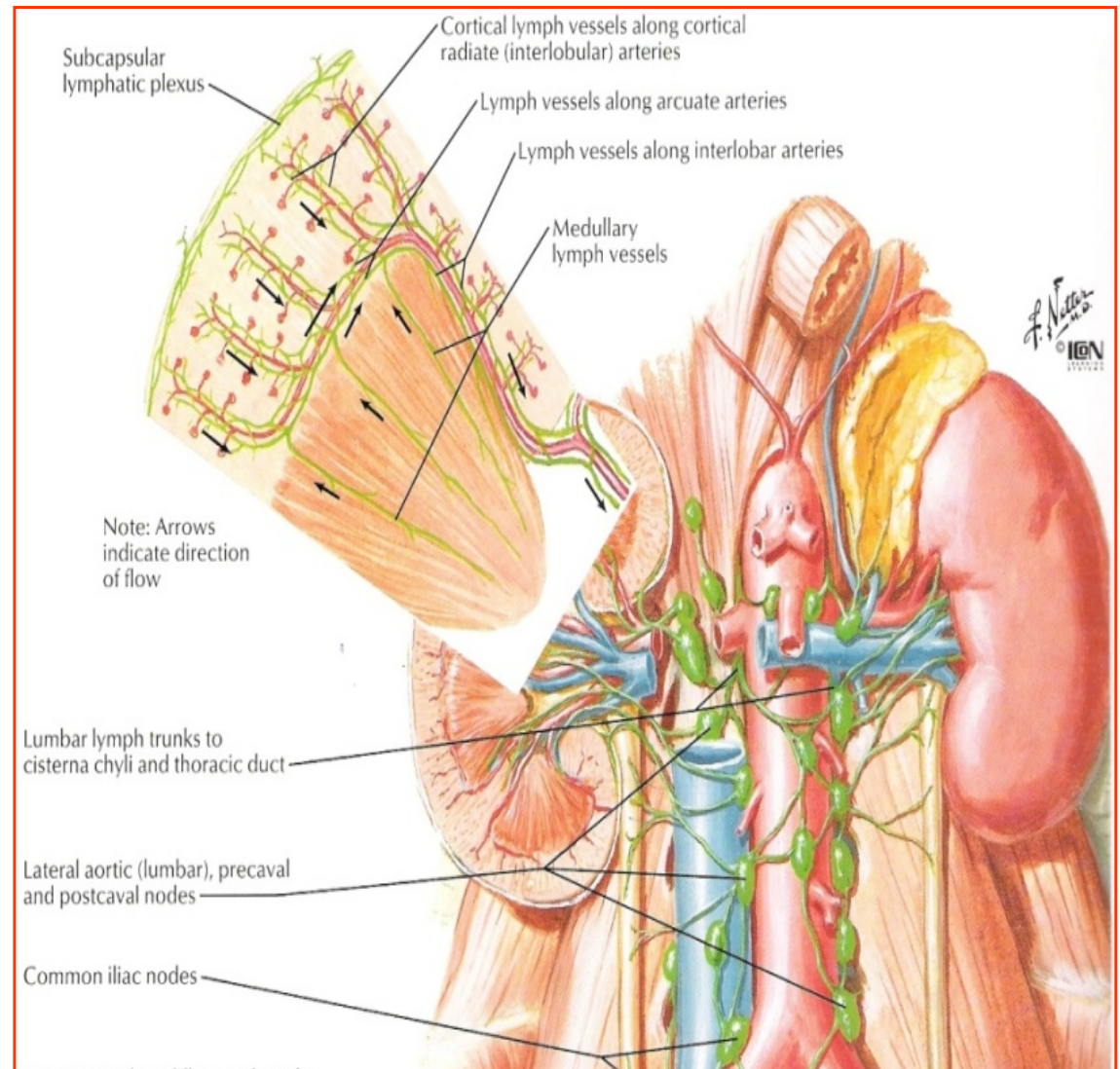
- Both renal veins drain to the inferior vena cava.
- The **left renal vein** is three times longer than the right (7.5 cm and 2.5 cm).
- So, for this reason the left kidney is the preferred side for live donor nephrectomy.
- It runs from its origin in the renal hilum, posterior to the **splenic vein** and the **body of pancreas**, and then across the **anterior to aorta**, just below the origin of the **superior mesenteric artery**.
- The left gonadal vein enters it from below and the left suprarenal vein, usually receiving one of the left inferior phrenic veins, enters it above but nearer the midline.
- The left renal vein enters the inferior vena cava a little above the right vein.



- The **right renal vein** is behind the 2nd part of the duodenum and sometimes behind the lateral part of the head of the pancreas

❑ Lymphatic Drainage:

- The lymph vessels follow the arteries.
- Lymph drains to the **lateral aortic lymph nodes** around the origin of the renal artery.



❑ Nerve Supply:

- The nerve supply is the renal sympathetic plexus. The afferent fibers that travel through the renal plexus enter the spinal cord in the 10th, 11th, and 12th thoracic nerves.

THANK YOU

